

Data_Scientist — Step 1

Learn Python & SQL

Detailed Learning Notes • CareerCompass

1. Learning Objective

Build the coding foundation used to acquire, transform, query and analyze data.

2. Core Concepts — Detailed Breakdown

Python

Learn variables, control flow, functions, modules, exceptions, file handling and environments, then move into NumPy, pandas and notebooks. Write reusable functions instead of one-off notebook cells.

SQL fundamentals

Understand SELECT, filtering, sorting, aggregation, GROUP BY, HAVING, joins, subqueries and common table expressions. SQL is often the first layer of analysis against structured business data.

Data types and schemas

Understand numeric, categorical, text, date/time and identifier fields. A correct query can still produce a wrong result if joins multiply rows or a field has an unexpected meaning.

Reproducibility

Keep queries, notebooks, assumptions and datasets organized. Record data sources and transformations so the analysis can be audited.

3. Recommended Learning Workflow

- Review Python basics.
- Practice pandas DataFrames and missing-data handling.
- Write SQL queries from simple to multi-table.
- Combine SQL extraction with pandas analysis.
- Build one notebook from raw data to conclusion.

5. Hands-on Project / Practice

Analyze a sales dataset stored in MySQL or SQLite. Write SQL queries for revenue by month/category/customer, export the result to pandas, calculate additional metrics, visualize trends and write five evidence-backed findings.

6. Common Mistakes & Pitfalls

- Using SELECT * everywhere.
- Joining tables without checking cardinality.
- Converting every SQL task into Python when the database can aggregate efficiently.
- Hard-coding assumptions without documentation.
- Confusing identifiers with measures.

7. Completion Checklist

- I can write multi-table SQL queries.
- I can manipulate DataFrames.
- I can identify data types and schema problems.
- I can move data between SQL and Python.
- I can make a reproducible analysis notebook.

8. Interview / Assessment Questions

- What is the difference between WHERE and HAVING?
- Why can a join duplicate rows?
- When should aggregation happen in SQL?
- What is a DataFrame?
- How do you handle missing values?

9. Recommended References

Python Tutorial: <https://docs.python.org/3/tutorial/>

pandas User Guide: https://pandas.pydata.org/docs/user_guide/

Data Science Tutorial: <https://www.youtube.com/watch?v=ua-CiDNNj30>

Data Analyst Bootcamp: <https://www.youtube.com/watch?v=PSNXoAs2FtQ>

Note: These notes are designed as a structured learning guide. Use the references for deeper, current documentation and hands-on practice.